

A Bounded-Generator Counterexample to Semigroup Little-Hölder Biduality

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Abstract

González-Pérez asked whether the little semigroup Hölder class $\lambda_\alpha(\mathcal{T})$ always has bidual isomorphic to the big class $\Lambda_\alpha(\mathcal{T})$. We give a negative answer for every $0 < \alpha < 1$, already for an ergodic symmetric Markov semigroup on the finite commutative von Neumann algebra $\ell_\infty(\mathbb{N})$. The generator is the bounded projection $A = I - E$, where E is expectation onto the constants. Consequently all elements are little Hölder and both classes equal ℓ_∞ with equivalent norms. Nevertheless ℓ_∞^{**} has strictly larger density character than ℓ_∞ , so the two spaces are not even abstractly isomorphic.

1 Source question

For a Markov semigroup $\mathcal{T} = (T_t)_{t \geq 0}$ with generator A and subordinated Poisson semigroup $P_s = e^{-sA^{1/2}}$, the source defines

$$\Lambda_\alpha(\mathcal{T}) = \left\{ f \in \mathcal{M} : \sup_{s>0} \left\| s^{1-\alpha} \frac{dP_s}{ds}(f) \right\|_\infty < \infty \right\},$$

$$\|f\|_{\Lambda_\alpha} = \max \left\{ \|f\|_\infty, \sup_{s>0} \left\| s^{1-\alpha} \frac{dP_s}{ds}(f) \right\|_\infty \right\}.$$

Open Problem P.3 on page 34 of the arXiv PDF (Section 8.1, page 3282 of the published version) defines

$$\lambda_\alpha(\mathcal{T}) = \left\{ f \in \Lambda_\alpha(\mathcal{T}) : \lim_{s \rightarrow 0^+} \left\| s^{1-\alpha} \frac{dP_s}{ds}(f) \right\|_\infty = 0 \right\}$$

and asks whether $\lambda_\alpha(\mathcal{T})^{**} \cong \Lambda_\alpha(\mathcal{T})$ [1].

(P.3) When (X, d) is a complete metric space, the classical Hölder classes over X are not just dual spaces but double duals. Its double predual is given by the *little Hölder classes*, i.e. the classes of those functions such that $|f(x) - f(y)| \in o(d(x, y)^\alpha)$, see [BCD87] and [Wea18 Chapter 4]. In our semigroup context there is a natural candidate to the double predual given by

$$\lambda_\alpha(\mathcal{T}) = \left\{ f \in \Lambda_\alpha(\mathcal{T}) : \lim_{s \rightarrow 0^+} \left\| s^{1-\alpha} \frac{dP_s}{ds}(f) \right\|_\infty = 0 \right\} \subset \Lambda_\alpha(\mathcal{T}).$$

It is a natural question whether $\lambda_\alpha(\mathcal{T})^{**} \cong \Lambda_\alpha(\mathcal{T})$

2 Proof intuition

The proposed little/big distinction only detects singular short-time behaviour of the generator. If $A^{1/2}$ is bounded, then $s^{1-\alpha} A^{1/2} P_s f \rightarrow 0$ uniformly for every f , so the little and big classes collapse to

the ambient von Neumann algebra. Choosing the ambient algebra to be ℓ_∞ turns this collapse into an obstruction: the desired biduality would assert that ℓ_∞^{**} is isomorphic to ℓ_∞ . A quotient-density argument rules out even an arbitrary Banach-space isomorphism. The expectation semigroup below is finite, commutative, symmetric, ergodic, and nonconstant, so the obstruction is not caused by the identity semigroup.

Theorem 1 (Counterexample to P.3). *For every $0 < \alpha < 1$ there are a finite commutative von Neumann algebra $(\mathcal{M}, \tau)$ and an ergodic symmetric Markov semigroup $\mathcal{T}$ on $\mathcal{M}$ such that*

$$\lambda_\alpha(\mathcal{T}) = \Lambda_\alpha(\mathcal{T}) = \mathcal{M}$$

with equivalent norms, but

$$\lambda_\alpha(\mathcal{T})^{**} \not\cong \Lambda_\alpha(\mathcal{T}).$$

Thus Open Problem P.3 has a negative answer in its stated generality.

3 Proof

Let $\mathcal{M} = \ell_\infty(\mathbb{N})$ and equip it with the faithful normal finite trace

$$\tau(f) = \sum_{n=1}^{\infty} 2^{-n} f_n, \quad f \in \mathcal{M}_+.$$

Since $\mathcal{M}$ is commutative, this normal state is a trace. Let

$$E(f) = \tau(f)\mathbf{1}$$

be the trace-preserving conditional expectation onto the constants and put

$$T_t = E + e^{-t}(I - E) = e^{-t}I + (1 - e^{-t})E, \quad t \geq 0. \quad (1)$$

The map E is normal, unital, completely positive, trace preserving, and an orthogonal projection on $L^2(\mathcal{M}, \tau)$. Hence every T_t is Markovian and self-adjoint on L^2 . Since $E^2 = E$, formula (1) also gives $T_t T_u = T_{t+u}$ and pointwise weak-* continuity. Its fixed-point algebra is $\mathbb{C}\mathbf{1}$, so the semigroup is ergodic. Its positive self-adjoint generator is

$$A = I - E.$$

This is an orthogonal projection, whence $A^{1/2} = A$. Therefore the Poisson semigroup equals T_s and

$$P_s = E + e^{-s}(I - E), \quad \frac{dP_s}{ds} = -e^{-s}(I - E). \quad (2)$$

Fix $0 < \alpha < 1$. It follows from (2) that for every $f \in \mathcal{M}$,

$$\sup_{s>0} \left\| s^{1-\alpha} \frac{dP_s}{ds}(f) \right\|_\infty = c_\alpha \|(I - E)f\|_\infty, \quad (3)$$

$$c_\alpha = \sup_{s>0} s^{1-\alpha} e^{-s} = (1 - \alpha)^{1-\alpha} e^{-(1-\alpha)}. \quad (4)$$

In particular, $\Lambda_\alpha(\mathcal{T}) = \mathcal{M}$. Moreover,

$$\lim_{s \rightarrow 0^+} s^{1-\alpha} e^{-s} \|(I - E)f\|_\infty = 0,$$

so every element is little Hölder and $\lambda_\alpha(\mathcal{T}) = \Lambda_\alpha(\mathcal{T}) = \mathcal{M}$. Their common inherited norm is

$$\|f\|_{\Lambda_\alpha} = \max\{\|f\|_\infty, c_\alpha \|(I - E)f\|_\infty\},$$

which is equivalent to $\|f\|_\infty$ because E is contractive and hence $\|(I - E)f\|_\infty \leq 2\|f\|_\infty$.

It remains to show that $\mathcal{M}^{**}$ is not isomorphic to $\mathcal{M}$. Write $\mathfrak{c} = 2^{\aleph_0}$. The set $\{0, 1\}^{\mathbb{N}} \subset \ell_\infty$ is 1-separated and has cardinal $\mathfrak{c}$, while $|\ell_\infty| = \mathfrak{c}$. Thus

$$\text{dens}(\mathcal{M}) = \mathfrak{c}. \quad (5)$$

We next exhibit a much larger quotient of $\mathcal{M}^{**}$. Identify $\ell_\infty = C(\beta\mathbb{N})$. The compactification $\beta\mathbb{N}$ contains a set I of $\mathfrak{c}$ distinct points. For completeness, choose a family of $\mathfrak{c}$ pairwise almost-disjoint infinite subsets of $\mathbb{N}$ (encode the finite initial segments of each infinite binary sequence) and extend each to a free ultrafilter; the resulting ultrafilters are distinct. The point masses $(\delta_p)_{p \in I}$ span an isometric copy

$$Y = \ell_1(I) \subset C(\beta\mathbb{N})^* = \mathcal{M}^*.$$

The restriction map

$$R : \mathcal{M}^{**} = (\mathcal{M}^*)^* \longrightarrow Y^* = \ell_\infty(I)$$

is surjective by Hahn–Banach. Density cannot increase under a continuous surjection. On the other hand, $\{0, 1\}^I \subset \ell_\infty(I)$ is 1-separated and has cardinal $2^\mathfrak{c}$. Consequently

$$\text{dens}(\mathcal{M}^{**}) \geq \text{dens}(\ell_\infty(I)) \geq 2^\mathfrak{c} > \mathfrak{c} = \text{dens}(\mathcal{M}). \quad (6)$$

Since density character is an isomorphic invariant, (5) and (6) imply $\mathcal{M}^{**} \not\cong \mathcal{M}$. Together with the identification of the two Hölder classes above, this proves the theorem.

4 Verification

From this packet directory, run

```
conda run --no-capture-output -n sandbox python \
  code/verify_expectation_semigroup.py
```

The script uses exact rational arithmetic on finite weighted analogues of E for $N = 2, 3, 5, 8, 16$. It verifies unitality, trace preservation, $L^2(\tau)$ self-adjointness, $E^2 = E$, $A^2 = A$, and the exact semigroup product formula for several rational parameters. It also reports the closed-form maximizing point $s = 1 - \alpha$ in (3). These checks are sanity tests for the explicit algebra; the infinite-dimensional density argument is the formal proof and is not replaced by computation. The recorded run ended with **all checks passed**; the complete output is stored in **verification_report.txt**.

5 Novelty bounds, limitations, and review recommendation

A bounded search on 19 July 2026 covered the run’s four lightweight indexes, the exact arXiv id and title, the exact P.3 formula and phrases “little Hölder”, “semigroup Hölder bidual”, and “bounded generator”, the published 2022 article, and later semigroup Campanato literature. The closest later item found was Hong–Jing’s 2024 paper [2], which answers a Campanato/Lipschitz equivalence problem but does not discuss P.3 or biduality. No source containing this counterexample or an answer to P.3 was found. Novelty confidence is moderate–high, subject to a full citation review; the construction is elementary enough that an uncatalogued observation remains possible.

The theorem refutes only the unrestricted formulation of P.3. A positive version may still hold under hypotheses forcing genuine short-time smoothing, or otherwise ensuring that $\Lambda_\alpha(\mathcal{T})$ is a proper subspace of $\Lambda_\alpha(\mathcal{T})$. It does not address P.1, P.2, P.4, or P.5.

Human review is recommended, with focus on the interpretation of the source’s Banach-space isomorphism, the density-quotient step, and whether an intended unstated hypothesis was meant to exclude bounded generators. The density argument deliberately proves failure of arbitrary isomorphism, not merely failure of the canonical bidual map.

References

- [1] A. González-Pérez, *Hölder classes via semigroups and Riesz transforms*, Math. Z. 301 (2022), 3245–3284; [arXiv:1904.10332](#).
- [2] G. Hong and Y. Jing, *Campanato spaces via quantum Markov semigroups on finite von Neumann algebras*, Int. Math. Res. Not. IMRN 2024, no. 21, 13611–13641; [arXiv:2409.14414](#).
- [3] N. Hindman and D. Strauss, *Algebra in the Stone–Čech Compactification*, 2nd ed., de Gruyter, 2012.